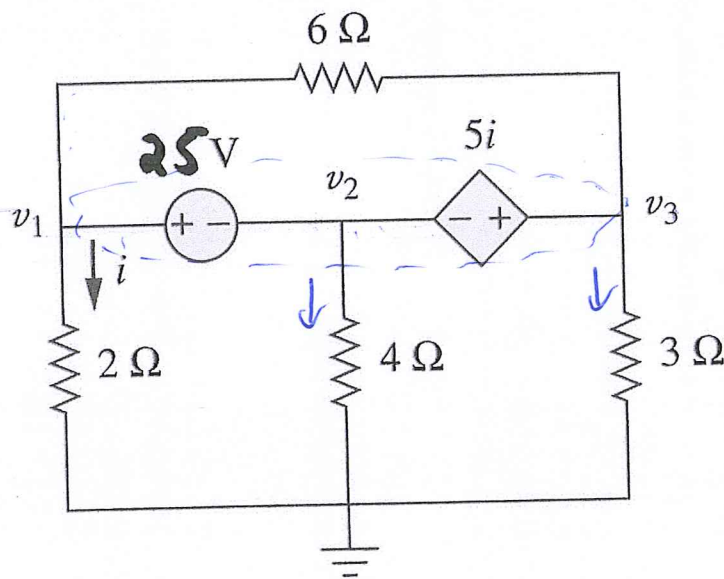


2) Find v_1 , v_2 , v_3 and i .



$$i' = \frac{v_1}{2}$$

$$v_1 - v_2 = 25 \quad \text{①}; \quad v_3 - v_2 = 5i'$$

$$v_3 - v_2 = \frac{5v_1}{2}$$

$$5v_1 + 2v_2 - 2v_3 = 0 \quad \text{②}$$

$$\frac{v_1}{2} + \frac{v_2}{4} + \frac{v_3}{3} = 0$$

$$\times 12: 6v_1 + 3v_2 + 4v_3 = 0 \quad \text{③}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 5 & 2 & -2 \\ 6 & 3 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 25 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 1 & -1 & 0 \\ 5 & 2 & -2 \\ 6 & 3 & 4 \\ 1 & -1 & 0 \\ 5 & 2 & -2 \end{bmatrix}$$

$$\begin{aligned} & (1)(2)(4) + 0 + 6(-1)(-2) \\ &= -50 - (-2)(3)(1) - (4)(-1)(5) \\ &= 8 + 0 + 12 - (-6) - (-20) \\ &= 46 \end{aligned}$$

$$\Delta_1 = \begin{bmatrix} 25 & -1 & 0 \\ 0 & 2 & -2 \\ 0 & 3 & 4 \\ 25 & -1 & 0 \\ 0 & 2 & -2 \end{bmatrix}$$

$$\begin{aligned} & (25)(2)(4) + 0 + 0 \\ &= -0 - (25)(3)(-2) - 0 \\ &= 200 - (-150) \\ &= 350 \end{aligned}$$

$$\Delta_2 = \begin{bmatrix} 1 & 25 & 0 \\ 5 & 0 & -2 \\ 6 & 0 & 4 \\ 1 & 25 & 0 \\ 5 & 0 & -2 \end{bmatrix}$$

$$\begin{aligned} & 0 + 0 + (6)(25)(-2) \\ &= -0 - 0 - (4)(25)(5) \\ &= -300 - 500 \\ &= -800 \end{aligned}$$

$$\Delta_3 = \begin{bmatrix} 1 & -1 & 25 \\ 5 & 2 & 0 \\ 6 & 3 & 0 \\ 1 & -1 & 25 \\ 5 & 2 & 0 \end{bmatrix}$$

$$\begin{aligned} & 0 + 5(3)(25) + 0 \\ &= - (25)(2)(6) - 0 - 0 \\ &= 375 - 300 \\ &= 75 \end{aligned}$$

$$V_1 = \frac{\Delta_1}{\Delta} = 7,609V; \quad V_2 = \frac{\Delta_2}{\Delta} = -17,391; \quad V_3 = \frac{\Delta_3}{\Delta} = 1,6304V$$

$$i = \frac{V_1}{2} = 3,805A$$